Progressive exercises

E1 · Direct application. We define on $\mathbb{R}$ the functions $f(x) = (2x-1)^5$ and $g(x) = e^{-x^2}$. Calculate $f'(x)$ and $g'(x)$.

E2 · Application and justification. Let f the function defined on $\mathbb{R}$ by $f(x) = x^4 - 6x^2$. Determine the intervals on which f is convex or concave.

E3 · Method to organise. Let $\mathcal{C}$ be the curve of the exponential function. Find an equation of its tangent at the point with x-coordinate 1. Using convexity, deduce an inequality valid for every real x.

E4 · Choice of strategy. Prove that the equation $x^3 + 2x = 5$ has a unique real solution. Bound this solution between two consecutive integers.

E5 · Synthesis or proof. A bisection algorithm starts from the interval $[0;2]$ and divides its length by two at each step. Determine the minimum number of steps guaranteeing an interval of length at most 10^{-3}.